

TANYA MISTRY

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EDUCATION

Northeastern University Master of Science in Computer Science	December 2026
<u>Coursework:</u> Database Management Systems, Algorithms, Cloud Computing, Data Mining	GPA: 3.67/4.0
University of Mumbai Bachelor of Technology in Information Technology and Minor in Data Science	June 2024
<u>Coursework:</u> Big Data Analysis, Machine Learning, Artificial Intelligence, Statistical Analysis	GPA: 3.7/4.0

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, R

Frameworks & Libraries: React, Express, Next.js, Node.js, Pandas, NumPy, Scikit-learn, PyTorch, OpenCV

Data & AI: Apache Spark, Kafka, PostgreSQL, pgvector, Snowflake, RAG, LLMs, Sentence-Transformers, ETL Pipelines

Cloud & DevOps: AWS (S3, Glue, Lambda, Athena), Docker, Git

Visualization: Tableau, Power BI, Streamlit, Plotly

PROFESSIONAL EXPERIENCE

Engineering Co-op | Northeastern University (EDGE) January 2026 - June 2026

- Engineered an **ETL pipeline** to migrate Coursera course content into Canvas LMS by extracting structured course data, reconstructing module hierarchies, and generating IMS Common Cartridge (IMSCC) packages, eliminating an estimated **10 hours/week** of manual migration work across the team
- Owned the end-to-end **design, development, testing, and deployment** of a Canvas LTI 1.3 platform using **Next.js, Express, and PostgreSQL**
- Developed a **FERPA-aligned analytics** layer tracking student page views, time-on-page via 30-second heartbeats, and component interactions, powering a course-level dashboard with per-student engagement insights
- Automated a **Word-to-QTI** conversion pipeline that generated Canvas-compatible quiz packages, eliminating an estimated **3+** hours of manual quiz creation per course

Data Analyst Intern | One Hand Clap June 2023 - May 2024

- Created automated reporting workflows using **ListenFirst API, SQL, and Excel** to track weekly engagement metrics for **Amazon Prime Video** that reduced manual effort across teams
- Developed interactive **Tableau** dashboards comparing Prime Video and **Netflix** by content type and user reach, which informed content strategy and contributed to an **8%** lift in engagement
- Analyzed competitor trends for **Bumble India** using **Python (Pandas, Matplotlib)** to identify popular content patterns to improve reporting clarity for non-technical teams
- Collaborated with **cross-functional teams** to communicate upcoming trends and strategies to boost Amazon Prime Video's user engagement

PROJECTS

TrialSage | Python, PostgreSQL, pgvector, Llama 3.1, Sentence-Transformers, Docker, Claude Code

- Built an **AI-powered hybrid Retrieval-Augmented Generation (RAG)** system over **39K+ clinical trials** and **966K+ eligibility criteria**, integrating **text-to-SQL** with **semantic vector search** for structured and free-text trial matching
- Designed an **LLM query routing pipeline** with confidence scoring and rule-based fallback to classify and route structured, semantic, and hybrid queries
- Engineered **polarity-aware NLP** to distinguish inclusion from exclusion criteria and prevent unsafe eligibility inversions, identifying and correcting **1,455 mislabeled criteria**
- Leveraged **Claude Code for AI-assisted software development**, debugging, and test generation; implemented **citation guardrails** with zero fabricated trial-ID citations, **200+ automated tests**, and latency/token tracing

Real-Time Crypto Market Data Pipeline | Python, Kafka, Spark, PostgreSQL, Docker

- Developed a real-time **streaming pipeline** ingesting live trades from Coinbase WebSocket, processing **10K+** events/minute through **Kafka** and **Spark** Structured Streaming
- Implemented OHLCV candle aggregation with watermarking for accurate **time-windowed analytics**, storing raw trades and 1-minute candles in **PostgreSQL**
- Developed an auto-refreshing **Streamlit** dashboard with **Plotly** candlestick charts for **real-time market visualization**
- Containerized entire stack with **Docker Compose** for reproducible deployment

Automated Chart Insights Generator | Python, OpenCV, EasyOCR, Ollama, Streamlit

- Created a **computer vision pipeline** that extracts axis labels and data points from chart images using **OpenCV** and **EasyOCR**, then passes structured metadata to a local **LLaVA** vision model to generate plain-English insights across multiple chart types
- Delivered a **Streamlit UI** with **4** categorized insight sections (summary, findings, trends, anomalies), **JSON export**, and interactive **chart analysis**